



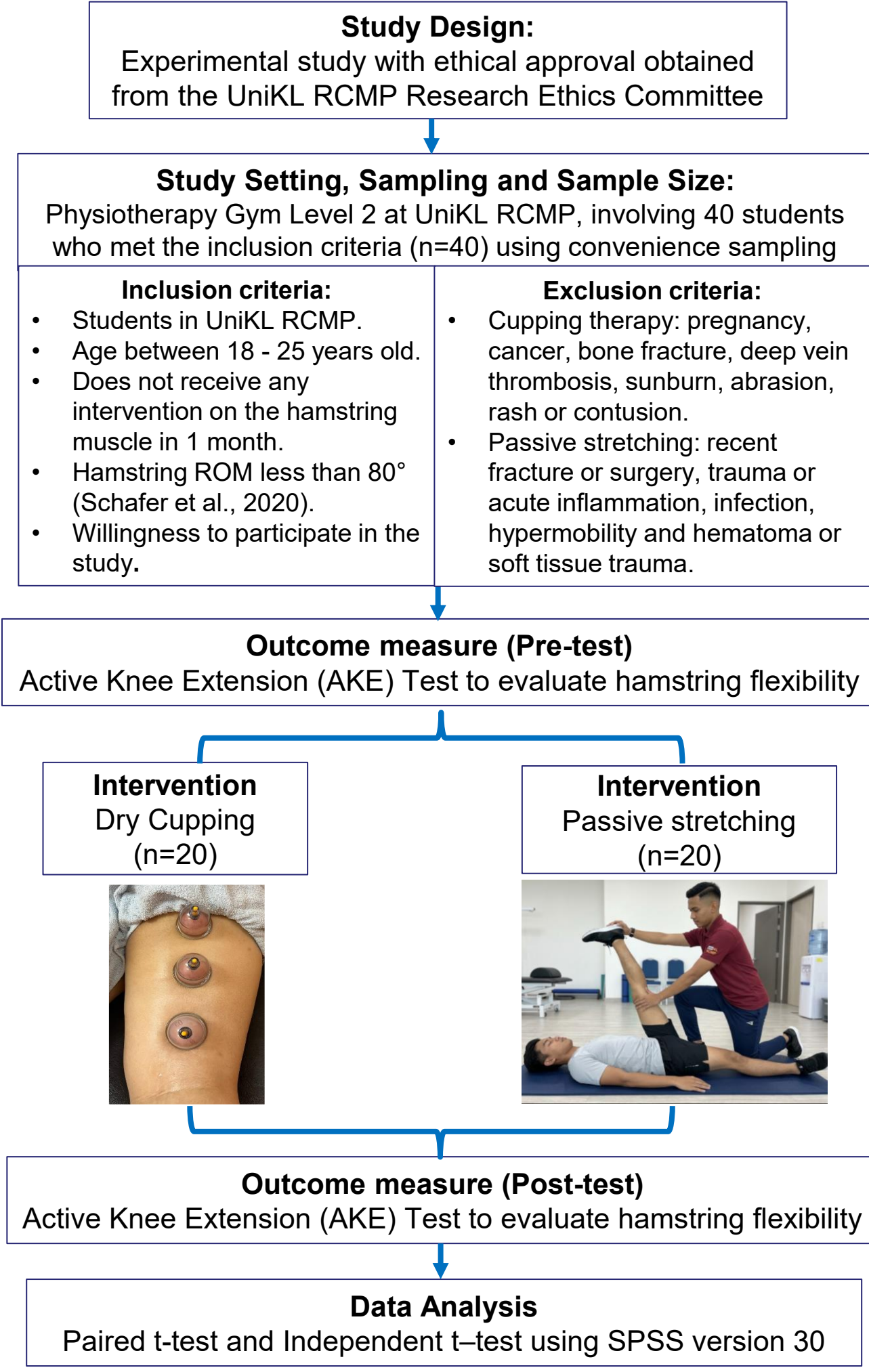
INTRODUCTION

- Hamstring tightness is a common among young adults and may reduce flexibility and increase the risk of musculoskeletal injury (Shaikh et al., 2023).
- University students frequently experience reduced hamstring flexibility, highlighting the need for effective flexibility interventions (Liyanage et al., 2024).
- Dry cupping therapy may improve local blood flow and muscle flexibility (Schultz, 2020), while passive stretching lengthens the muscle to improve flexibility (Morcelli et al., 2013).
- Although both interventions are used to improve muscle flexibility, evidence directly comparing their immediate effects on hamstring flexibility among university students remains limited.

 **Objective:** To investigate and compare the immediate effects of dry cupping therapy and passive stretching on hamstring flexibility among students at UniKL RCMP.



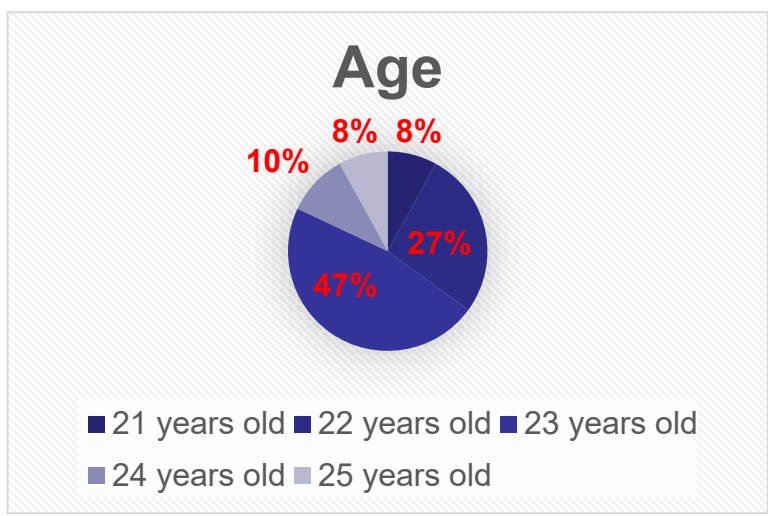
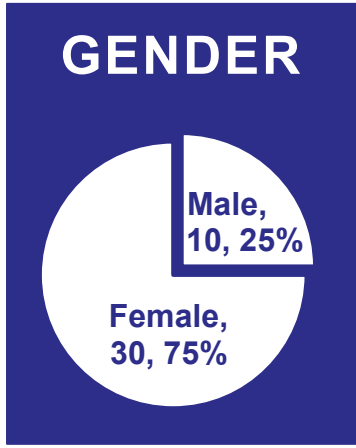
METHODOLOGY



RESULTS



Demographic Data



The Immediate Effect of Dry Cupping Therapy and Passive Stretching on Hamstring Flexibility among Student at UniKL RCMP by using AKE Test (n=40)

Variables	Pre - AKE Mean ± SD	Post - AKE Mean ± SD	P - value
Dry Cupping	40.56 ± 8.51	53.01 ± 9.04	< 0.001*
Passive Stretching	45.19 ± 11.24	58.11 ± 10.10	< 0.001*

*p < 0.001 is significantly difference, paired t-test.



Comparison of Dry Cupping Therapy and Passive Stretching on Hamstring Flexibility among Student at UniKI RCMP (n=40)

Variables	Dry Cupping	Passive Stretching	P - value
Pre - AKE (Mean ± SD)	40.56 ± 8.51	45.19 ± 11.24	0.150
Post - AKE (Mean ± SD)	53.01 ± 9.04	58.11 ± 10.10	0.106

*p < 0.05 is significantly difference, independent t-test.

The results showed that both interventions independently led to significant improvements in hamstring flexibility. However, there was no significant difference in the effectiveness when comparing the two interventions directly.



DISCUSSION

- Equally effective tightness relief:** Both interventions successfully reduce hamstring stiffness and restore range of motion in acute settings (Bhimani et al. 2019).
- Dry cupping** enhances local blood flow, reduces muscle tightness and promotes relaxation which contributed to the increase of range of motion (ROM) (Schultz, 2020).
- Passive stretching** improves flexibility through reduced muscle-tendon unit stiffness and increased stretch tolerance (Morcelli et al., 2013).
- Although both methods are equally effective, the study underscores the potential of these noninvasive interventions for flexibility training and rehabilitation, offering individuals multiple viable options to improve flexibility.
- Both approaches may be considered for immediate improvements in hamstring flexibility. Future studies with larger samples, randomized controlled designs and longer follow-up are recommended to confirm these findings.



ACKNOWLEDGEMENT

Universiti Kuala Lumpur Royal College of Medicine Perak (UniKL RCMP) for providing clinical facilities and research support. Participants: UniKL RCMP student volunteers for their participation.



DECLARATION OF CONFLICT FOR ALL AUTHORS

The authors declare that they have no conflict of interest.



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